

# Lecture 25: DNA, Genetics & Gene Regulation

BINF 4002 — Machine Learning for Health

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April 21, 2026

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## Modality progression:

- L17–18: EHR & Claims    • L19–20: Clinical Text & NLP
- L21–22: Clinical Imaging    • L23: Population Health    • L24: Fairness
- L25: DNA, Genetics & Gene Regulation ← **today**
- L26: Proteins, Molecules & Structural Biology
- L27: Modern Biological AI

## Three lectures on molecular/biological data:

1. **Today:** The genome — DNA, genetic variation, association analysis, gene regulation

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1. **Today:** The genome — DNA, genetic variation, association analysis, gene regulation
2. **L26:** The molecular level — proteins, small molecules, classical analysis methods
3. **L27:** What deep learning adds — and where it fails

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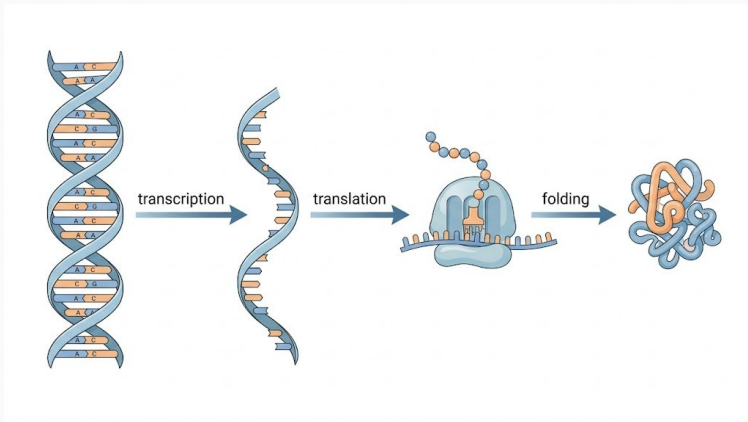
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6. **Define** PRS and explain the cross-ancestry transferability problem
7. **Explain** how gene expression connects genotype to phenotype, and why context matters

# DNA and the Central Dogma

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# The Central Dogma



Each arrow is a place where ML can help: variant effect prediction, expression prediction, structure prediction, function prediction. Today we focus on the first step: DNA and genetic variation.

# DNA Basics

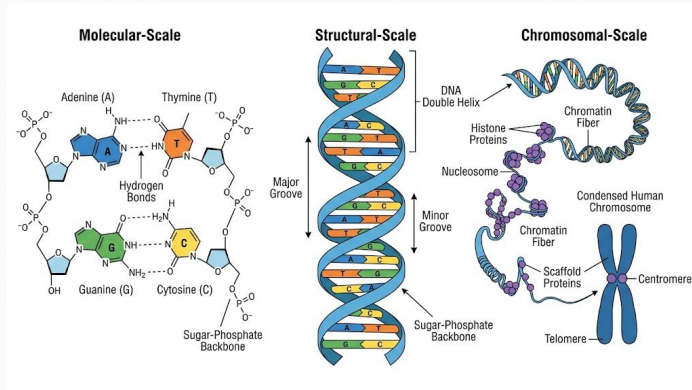
**Four bases:** A, T, G, C. Double helix with complementary base pairing (A-T, G-C).

**Scale:** 3.2 billion base pairs, 23 chromosome pairs, ~20,000 protein-coding genes — but protein-coding exons are only ~1–2% of the genome. Most DNA is noncoding; some is regulatory, and much remains incompletely characterized.

**The genome as a reference:** We have a “reference genome” (GRCh38). Individual variation is measured *relative to* this reference.

| What we have                                | Scale                                  |
|---------------------------------------------|----------------------------------------|
| Human genomes genotyped (arrays)            | >1 million (e.g., UK Biobank, 23andMe) |
| Whole-genome sequences (WGS)                | ~76,000 (gnomAD v4)                    |
| Exome sequences (WES)                       | ~730,000 (gnomAD v4)                   |
| Protein-coding genes identified             | ~20,000                                |
| Variant–disease associations (GWAS Catalog) | ~400,000                               |
| Variants with functional characterization   | Much less                              |

# DNA at Three Scales



**Left:** Base pairing chemistry (A-T, G-C) on the sugar-phosphate backbone. **Center:** The double helix with major/minor grooves (where proteins bind to regulate gene expression). **Right:** DNA wraps around histones → chromatin fiber → condensed chromosome. Packaging affects which genes are accessible.

# Genetic Variants

**SNPs** (single nucleotide polymorphisms): one base differs from reference. The most common variant type ( $\sim 4\text{--}5\text{M}$  per individual).

Other types: insertions/deletions (indels), structural variants (large rearrangements). Today: SNPs — the workhorse of genetic association studies.

**Minor allele frequency (MAF):** Common ( $> 1\%$ , well-tagged by arrays) vs. rare ( $< 1\%$ , require sequencing).

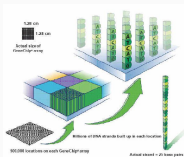
**The genotype matrix:**  $n$  individuals  $\times$   $p$  SNPs. Each entry  $\in \{0, 1, 2\}$  = copies of alternate allele.

## Data Type Box

**TABULAR**, but  $p \gg n$  ( $\sim 5\text{M}$  SNPs,  $\sim 10^3\text{--}10^6$  individuals), with structured correlation (LD) and population stratification as dominant confounder.

# Measuring Genetic Variation

## SNP Arrays



Targeted: ~500K–5M pre-selected SNPs. Cheap (~\$50–\$200). Most large GWAS use arrays + imputation. Miss rare variants entirely.

**Clinical applications:** BRCA1/2 testing, pharmacogenomics, carrier screening. **Key databases:** gnomAD, ClinVar, GWAS Catalog, 1000 Genomes.

## Exome Sequencing (WES)

Sequences protein-coding regions (~2% of genome). Better coding coverage than arrays, better depth than WGS. Moderate cost (~\$100–\$400). Widely used in clinical diagnostics and large cohorts (gnomAD: ~730K exomes). Misses noncoding regulatory variants.

## Whole-Genome Seq.

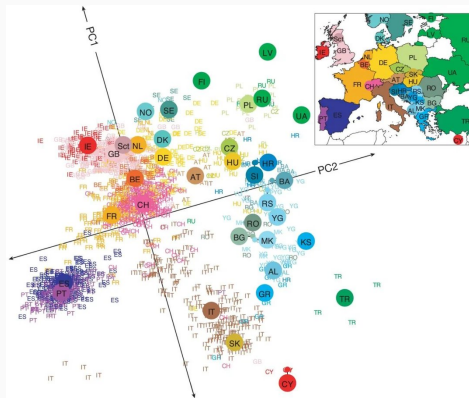


Comprehensive: all 3.2B positions. More expensive (~\$200–\$1,000). Detects rare variants, noncoding, SVs. Growing but still smaller cohorts (gnomAD: ~76K WGS).

# Population Structure

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# Genes Mirror Geography



Novembre et al., *Nature* 2008. PCA of  $\sim 200K$  SNPs from  $\sim 1,400$  Europeans recovers the map of Europe. Each point is a person; position is determined entirely by genetics. **The dominant signal in genetic data is ancestry, not disease.**

# Why This Is a Problem for ML

**You just saw:** PCA on genetic data recovers geography. Human migration history imprinted allele-frequency differences across populations over thousands of generations.

**Disease effects** are tiny: common-variant per-allele effects usually have odds ratios around 1.01–1.2. Collectively, thousands of such variants explain 5–50% of trait variance.

## Consequence for ML

A naive classifier on raw genotype features will learn *ancestry* and call it “disease risk.” Any ancestry group with higher prevalence (even from socioeconomic factors) gets flagged as high-risk.

**Connection to L17:** An extreme  $p \gg n$  problem. With  $\sim 5\text{M}$  features and  $\sim 5\text{K}$  samples, overfitting to population structure is the default.

# Linkage Disequilibrium (LD)

**LD:** Nearby variants on the same chromosome are co-inherited. They are *not* independent features.

## Consequences for ML:

1. A significant GWAS hit may not be the causal variant — it could be a correlated neighbor

**Analogy:** LD is to genetics what multicollinearity is to regression — correlated features where the model can't distinguish cause from proxy.

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3. Regularized models (LASSO) can arbitrarily choose one SNP from an LD block

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# GWAS

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# Genome-Wide Association Study: The Approach

**Goal:** Find SNPs *associated* with a trait (disease status, biomarker, etc.)

**Method:** Test each SNP *independently* ( $\chi^2$  or logistic regression for binary traits; linear regression for continuous).

**Key distinction from ML:** GWAS asks “is  $\hat{\beta}$  significantly non-zero?” — not “does this SNP improve prediction?” We identify *which* variants matter, not build the best classifier.

**Why not a multivariate model?**  $p \gg n$ : can't fit 5M coefficients with 5K samples. Each SNP has a tiny effect (OR  $\approx$  1.01–1.2). GWAS is *feature screening*, not prediction.

**Generalizes beyond disease:** Replace the trait with gene expression  $\rightarrow$  eQTL; chromatin accessibility  $\rightarrow$  caQTL. Same framework, different response.

## Effect Sizes: What “Significant” Means

**GWAS reports effect sizes** ( $\hat{\beta}$  or odds ratio), not prediction accuracy.

- Typical genome-wide significant SNP: OR  $\approx$  1.05–1.2, occasionally larger for specific traits

**Why small effects matter:** Disease is polygenic — thousands of small effects accumulate. Finding the right 50 SNPs out of 5M is a needle-in-a-haystack problem. Statistical significance is the tool for the search, even when each individual effect is tiny.

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- Significance tells you a variant is *associated*; it says nothing about how well you can *predict* the trait

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**GWAS is not just an academic exercise.** Real discoveries have changed clinical practice:

- **PCSK9** → **cholesterol drugs**: GWAS found PCSK9 variants → rare loss-of-function mutations showed lower heart attack risk → FDA-approved antibodies (evolocumab, alirocumab) lower LDL by ~60%.

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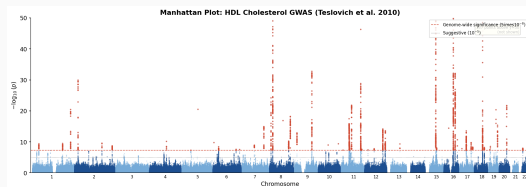
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- **HLA** → **autoimmune disease**: Strongest GWAS signals for T1D, RA, and celiac all map to HLA/MHC on chr6 — pointing to immune mechanisms.

# Multiple Testing: 5 Million Hypotheses

Under the null, p-values  $\sim \text{Uniform}(0, 1)$ . At  $\alpha = 0.05$  with  $5 \times 10^6$  tests: 250,000 expected false positives.

**Bonferroni:**  $\alpha_{\text{genome-wide}} = 0.05/m$ . Convention:  $5 \times 10^{-8}$  ( $\sim 1\text{M}$  effective tests). BH/FDR is less conservative; standard in eQTL analysis.



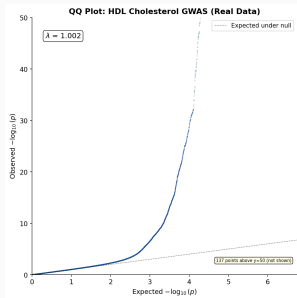
[Notebook Figure 1] Real GWAS data: HDL cholesterol (Teslovich et al., 2010). Peaks above the red line = genome-wide significant loci.

# QQ Plot and Genomic Inflation

**QQ plot:** Expected vs. observed  $-\log_{10}(p)$ . Since  $p \sim \text{Uniform}(0, 1)$  under the null, expected quantiles are known. **Genomic inflation**  $\lambda$ : median observed  $\chi^2 / 0.456$ .

## Reading the plot:

- Bulk follows diagonal  $\Rightarrow$  well-calibrated ( $\lambda \approx 1$ )

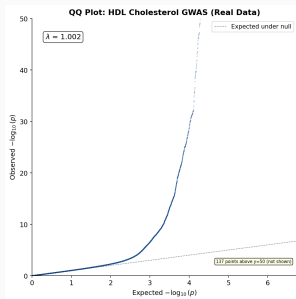


[Notebook Figure 2] Real data: HDL cholesterol GWAS.

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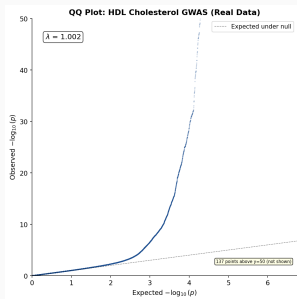
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- If the *whole curve* deviates upward ( $\lambda > 1$ ), suspect confounding or polygenicity

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# Correcting Population Stratification

**Problem:** Population stratification inflates test statistics genome-wide.

**Solution:** Include top PCs of the genotype matrix as covariates:

$$\text{logit}(P(\text{disease})) = \beta_{\text{SNP}} \cdot g + \sum_{k=1}^K \gamma_k \cdot \text{PC}_k$$

Typically  $K = 10\text{--}20$  PCs. These absorb ancestry signal (recall: PCA recovers the map of Europe), leaving genuine associations.

**Intuition:** If a SNP is associated with disease only because it tracks ancestry, adding PCs absorbs that signal  $\rightarrow$  non-significant. Genuine causal SNPs remain significant.

## PCs fix stratification, not LD

If a causal SNP is significant, its LD neighbors will *also* remain significant after PC adjustment — PCs do not break linkage. LD is addressed separately: clumping/pruning in PRS construction, fine-mapping for causal inference.

# Polygenic Risk Scores

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**After GWAS:** We have per-SNP effect sizes  $\hat{\beta}_i$ .

**PRS** for a new individual:  $\text{PRS} = \sum_{i=1}^M \hat{\beta}_i \times g_i, \quad g_i \in \{0, 1, 2\}.$

A **linear model with pre-computed coefficients** — no fitting on the target sample.

**The construction problem:** GWAS marginal  $\hat{\beta}$ s are correlated via LD. Naively summing them double-counts shared signal.

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- **C+T:** Keep most significant SNP per region; prune by  $r^2$  threshold. Simple but discards information.
- **Bayesian shrinkage (LDpred, PRS-CS):** Explicitly model LD and regularize  $\hat{\beta}$ s — same intuition as ridge regression. Better but more complex.

# PRS Transferability and Equity

~80% of GWAS participants are of European ancestry.

PRS trained on EUR data performs **substantially worse** in non-EUR populations:

| Ancestry group                 | Relative PRS performance |
|--------------------------------|--------------------------|
| European (training population) | 100% (reference)         |
| East Asian                     | ~50–60%                  |
| South Asian                    | ~40–60%                  |
| African                        | ~20–40%                  |

Illustrative summary based on Martin et al., *Nat Genetics* 2019. Performance varies substantially by trait and study design; these are approximate ranges across multiple phenotypes.

**Why:** LD patterns and allele frequencies differ across ancestries  $\Rightarrow$  the “right” tag SNP changes. **Clinical consequence:** EUR-derived PRS systematically underperforms for non-EUR patients. A health equity problem.

# Gene Expression and Regulation

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# From Genotype to Phenotype: The Missing Middle

**GWAS identifies associated SNPs.** But *how* do variants cause disease?



Most GWAS hits are in *non-coding* regions — they don't change the protein directly. Instead, they alter **gene regulation**: how much of a gene is transcribed into RNA.

**Example:** The FTO obesity locus doesn't affect FTO itself. The causal variant alters a long-range enhancer that controls IRX3, a gene 500kb away. Regulatory connections can span enormous genomic distances.

# How Gene Regulation Works

If most GWAS hits are in noncoding regions, what are they disrupting?

**Regulatory elements** control when, where, and how much a gene is transcribed:

- **Promoter:** DNA region just upstream of a gene where RNA polymerase binds to initiate transcription

**Why this matters:** A GWAS-significant SNP in an enhancer may alter TF binding → change expression of a distal gene → alter protein levels → shift disease risk. The variant doesn't change the protein — it changes *how much* protein is made, and *in which tissues*.

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- **Chromatin accessibility:** DNA wrapped tightly around histones is inaccessible; regulatory elements must be in “open chromatin” to function

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# Gene Expression: RNA as the Intermediate

**mRNA** = the intermediate product of the central dogma. Measuring mRNA tells you which genes are “active” in a given cell or tissue.

**RNA-seq:** Sequence all mRNA in a sample → a vector of expression levels across  $\sim 20,000$  genes. Each sample = a high-dimensional continuous measurement.

## Data Type Box

**HIGH-DIMENSIONAL CONTINUOUS.** Genes  $\times$  samples matrix ( $\sim 20K \times n$ ). For bulk RNA-seq: batch effects, library-size normalization, and the mean-variance relationship are the dominant data quality issues. For single-cell RNA-seq: sparsity and dropout add substantial additional complexity.

**Key resource:** GTEx v8 (Genotype-Tissue Expression) —  $\sim 17,000$  RNA-seq samples across 54 tissue sites from  $\sim 840$  donors, with matched genotypes. Enables tissue-specific eQTL analysis.

## eQTL: GWAS for Gene Expression

**Same statistical framework as GWAS**, but the response variable is gene expression instead of disease status.

**eQTL** (expression quantitative trait locus): a genetic variant that is statistically associated with the expression level of a gene.

- Test: for each SNP  $\times$  gene pair, is  $\hat{\beta}$  significantly non-zero?

**Extends to other phenotypes:** caQTL (chromatin accessibility), pQTL (protein levels), sQTL (splicing). Same framework, different molecular readout.

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- Identifies the regulatory *mechanism* behind GWAS hits

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# Why Context Matters: DNA Never Exists in a Vacuum

**The same variant can have different eQTL effects** in different tissues, cell types, and developmental stages.

A SNP that alters liver gene expression may have *no effect* in the brain — because the regulatory elements it disrupts are only active in liver.

## Implications for modeling

The *same* DNA in a liver cell and a neuron produces radically different outcomes. Gene regulation is **context-dependent**. DNA sequence alone does not fully determine expression across tissues, cell states, and environments. This is why DNA foundation models trained on sequence alone (L27) face fundamental limits.

## Wrap-up

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These are the resources your models will train on, validate against, and be compared to. Knowing what each contains — and what it *doesn't* — is essential.

## Key Takeaways

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4. **But GWAS discoveries have real clinical impact:** PCSK9 inhibitors, pharmacogenomic dosing, and autoimmune disease mechanisms all trace back to GWAS hits.

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6. **Context matters.** The same variant has different effects in different tissues — this limits what sequence-only models can learn.

1. Manhattan plot (HDL cholesterol GWAS, Teslovich et al. 2010)
2. QQ plot (same study — real test statistic calibration)
3. Gene expression heatmap (GTEx v8 — real tissue-specific patterns)

**How to use:** Run all cells. The notebook downloads data from the EBI GWAS Catalog and GTEx Portal. Explore: which chromosomes have the most lipid-associated loci? Which genes are most tissue-specific?

**Thursday (L26):** Proteins, molecules, and their classical analysis methods — MSA, conservation, fingerprints, docking. Then **L27:** modern biological AI.

**Questions?**

## Appendix: Multiple Testing Correction Methods

Three approaches to the same problem:

|                    | Bonferroni         | BH / FDR                | Permutation (min- $t$ / max- $p$ ) |
|--------------------|--------------------|-------------------------|------------------------------------|
| <b>Controls</b>    | FWER (any FP)      | FDR (proportion of FPs) | FWER (exact)                       |
| <b>Threshold</b>   | $\alpha/m$ (fixed) | Adaptive                | Empirical (from null dist.)        |
| <b>Correlation</b> | Ignores it         | Ignores it (less so)    | Accounts for LD                    |
| <b>Power</b>       | Lowest             | Moderate                | Highest for FWER                   |
| <b>Cost</b>        | Trivial            | Trivial                 | Expensive ( $10^3$ – $10^4$ perms) |
| <b>Standard in</b> | GWAS               | eQTL, diff. expression  | Small studies, imaging             |

**Why GWAS uses Bonferroni:** A false positive triggers expensive follow-up. The  $5 \times 10^{-8}$  convention already accounts for LD ( $\sim 1$ M effective tests, not 5M). **When permutation helps:** When tests are correlated (SNPs in LD, voxels in imaging), Bonferroni over-corrects. Westfall–Young / min- $t$  / max- $p$  estimate the null by shuffling labels, preserving correlation. Exact FWER with higher power, but at  $10^3$ – $10^4 \times$  computational cost.